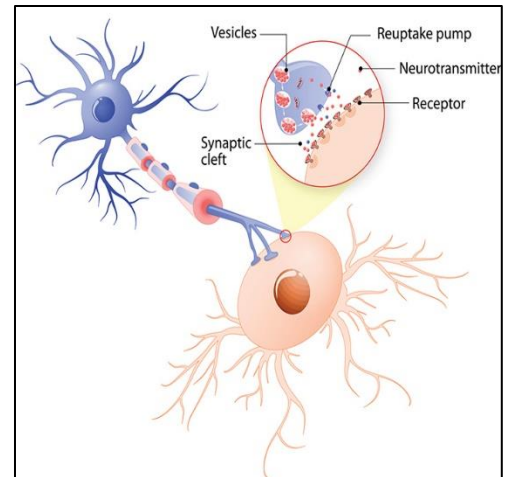


## 2. Neurotransmitters

### ➤ What are the neurotransmitters?

- are the brain chemicals that communicate information throughout our brain and body.
- They are released from synaptic vesicles in synapses into the synaptic cleft, where they are received by neurotransmitter receptors on the target cell.



### ➤ Classification of neurotransmitters

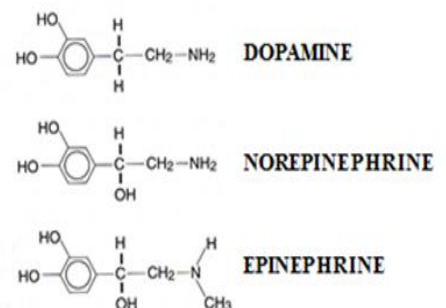
#### a) According to biochemical structure:

1. **Monoamines**: such as dopamine, noradrenaline, adrenaline, histamine, serotonin
2. **Amino acids**: such as glutamate, GABA (gamma-aminobutyric acid), glycine, aspartate
3. **Peptides**: such as opioids, endorphins, somatostatin, oxytocin, vasopressin
4. **Other**: such as acetylcholine (A.Ch), adenosine, nitric oxide

#### Note:

Dopamine, norepinephrine and epinephrine known as catecholamines and all are derived from tyrosine amino acid

#### Catecholamines



#### b) According to function:

- **Excitatory**: neurotransmitters increase electrical excitability on the post-synaptic side
- **Inhibitory**: neurotransmitters decrease electrical excitability on the post-synaptic side
- **Neuromodulators**: function to alter the strength of transmission between neurons by affecting the amount of neurotransmitter that is produced and released
- **Properties of neurotransmitters**:
  1. Synthesized in the presynaptic neuron.
  2. Localized to vesicles in the presynaptic neuron.
  3. Released from the presynaptic neuron under physiological conditions.
  4. Rapidly removed from the presynaptic cleft by uptake or degradation.
  5. Presence of receptor on the post-synaptic neuron.
  6. Binding to the receptor elicits biological response.

➤ **Types of neurotransmitter receptors**

The receptors that are activated by neurotransmitters are classified into two broad classes:

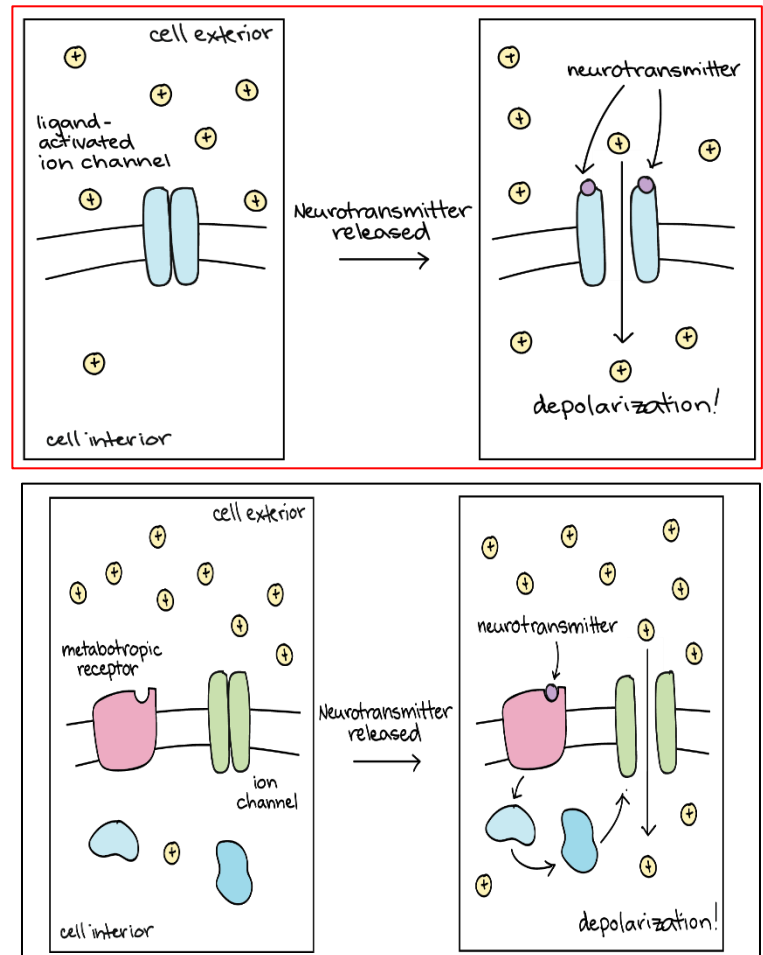
- 1- Ligand-activated ion channels**: These receptors are membrane-spanning ion channel proteins that open directly in response to ligand binding.
- 2- Metabotropic receptors**: These receptors are not themselves ion channels. Neurotransmitter binding triggers a signaling pathway, which may indirectly open or close channels (or have some other effect entirely).

➤ **Ligand-activated ion channels:**

- ✓ Also known as ionotropic receptors.
- ✓ They undergo a change in shape when neurotransmitter binds, causing the channel to open.
- ✓ This may have either an excitatory or an inhibitory effect

➤ **Metabotropic receptors**

- ✓ It affects ion channel opening and closing indirectly.
- ✓ Signaling through these metabotropic receptors depends on the activation of several molecules inside the cell and often involves a second messenger pathway.
- ✓ Because it involves more steps, signaling through metabotropic receptors is much slower than signaling through ligand-activated ion channels.



**Peripheral neurotransmitters**

- In the Autonomic nervous system (ANS), the major neurotransmitters are noradrenaline and adrenaline, which are released onto target organs and tissues by the sympathetic nervous system, and Acetyl Choline (ACh), which is released by the parasympathetic nervous system.

**Central neurotransmitters**

- In the Central nervous system (CNS), the major excitatory and inhibitory neurotransmitters are both amino acids, glutamate and  $\gamma$ -amino butyric

acid (GABA), respectively. These are the main mediators of fast signaling in the brain and spinal cord.

➤ **Neurotransmitter synthesis**

- The small ‘classical’ neurotransmitters (amino acids and amines) are manufactured by enzymes present in the presynaptic terminal.
- For example, the neurotransmitter glutamate is present in large amounts in the cell. In neurons, it is synthesized either from glucose provided by the Krebs cycle or from glutamine produced by glial cells.
- Peptide transmitters are manufactured in the cell body by ribosomes on the endoplasmic reticulum, and are processed and packaged via the extensive Golgi system into vesicles that are transported down the axon.

➤ **Neurotransmitter Inactivation**

- Acetyl Choline seems to be unique, in that it is inactivated by enzymatic degradation within the cleft itself. This is carried out by acetylcholinesterases (AChEs) in the synaptic cleft at the neuromuscular junction or other cholinergic synapses.
- The esterases cleave ACh into its inactive components, *choline* and *acetate*.
- There are three types of AChEs;<sup>1</sup> the membrane- bound AChEs, two soluble esterase enzymes. <sup>2</sup> One AChE found in the cerebrospinal fluid, and <sup>3</sup> the other, called *butyrylcholinesterase* or *pseudocholinesterase*, is found in plasma.
- Simple diffusion away from the synaptic cleft may be important in the inactivation of transported molecules, such as glutamate and GABA, and is the main mechanism of inactivation for peptide neurotransmitters.
- As diffusion may be slow, this may explain why peptide actions are so long-lasting.

- Peptides may also be removed by nonspecific peptidases and receptor-mediated endocytosis.

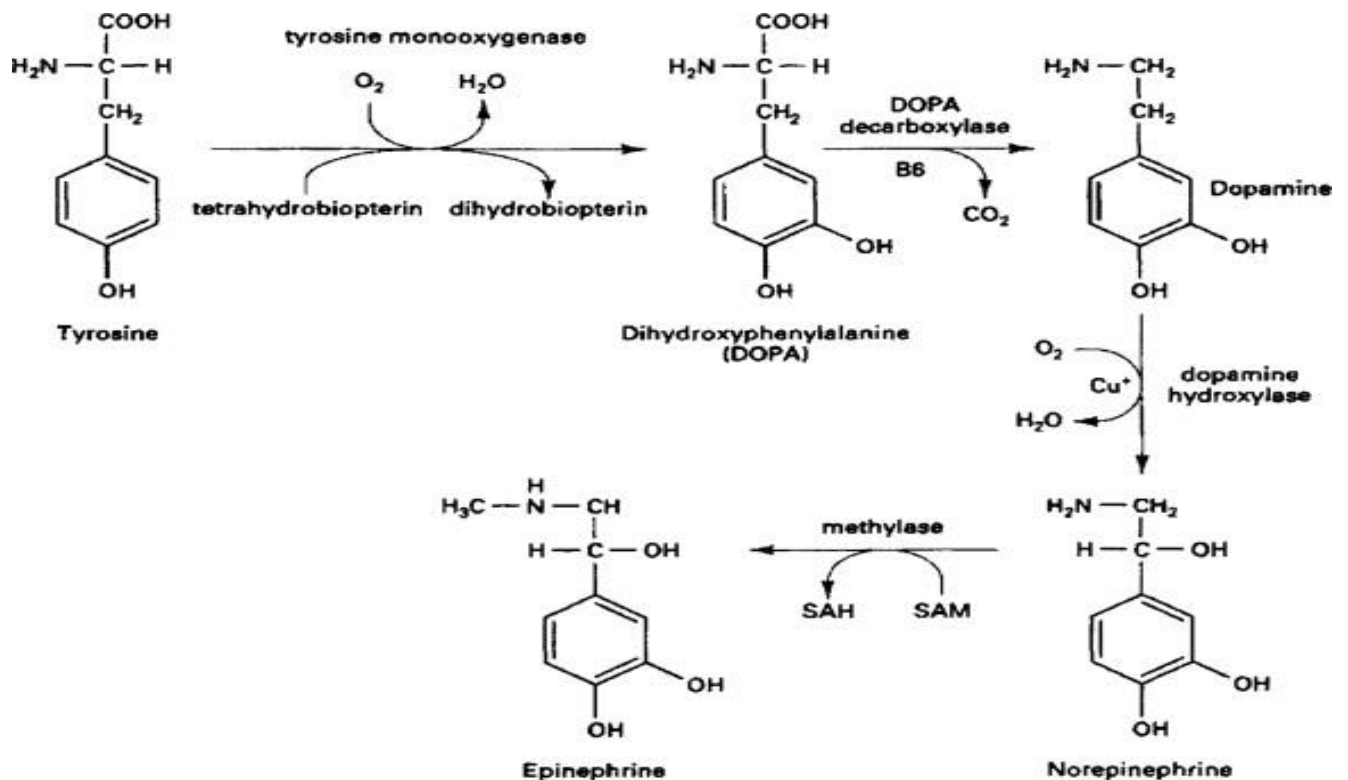
➤ **Catecholamines**

- Dopamine, norepinephrine, and epinephrine are biologically active (biogenic) amines that are collectively termed catecholamines.
- Dopamine and norepinephrine are synthesized in the brain and function as neurotransmitters.
- Norepinephrine is also synthesized in the adrenal medulla.

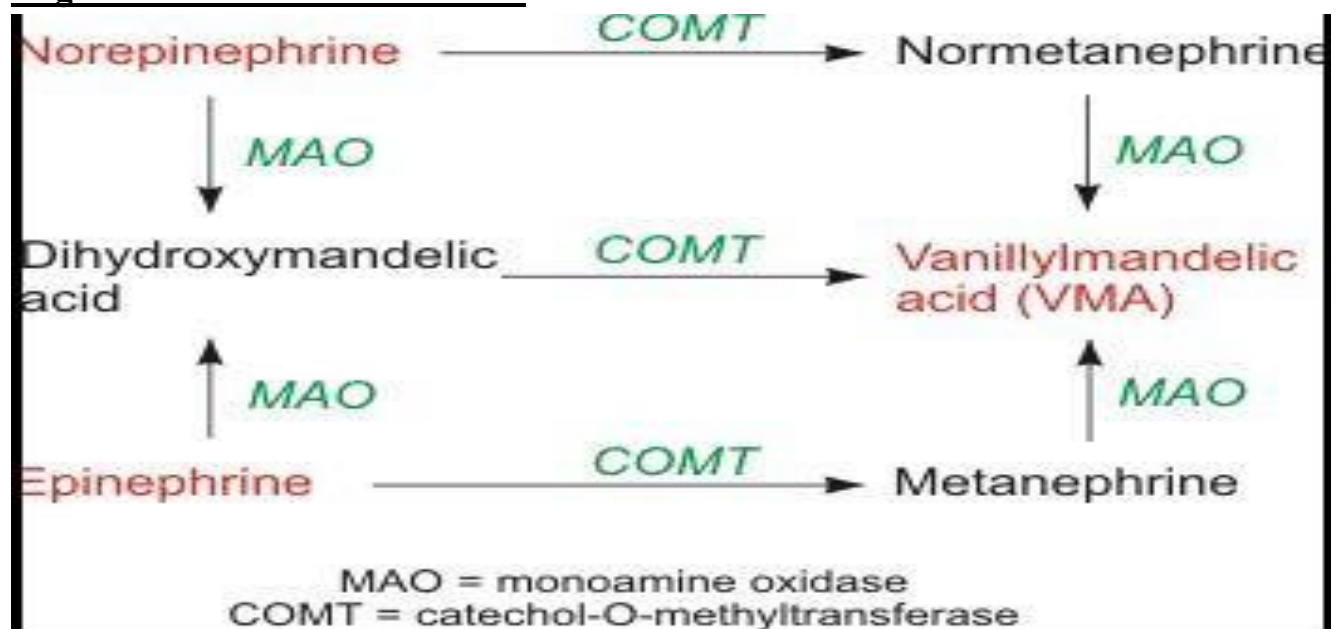
**Functions of Catecholamines**

- Both norepinephrine and epinephrine, are hormone regulators of carbohydrate and lipid metabolism.
- Norepinephrine and epinephrine are released from storage vesicles in the adrenal medulla in response to fight, exercise, cold, and low levels of blood glucose.
- They increase the degradation of glycogen and triacylglycerol, as well as increase blood pressure and the output of the heart. These effects are part of a coordinated response to prepare the individual for stress, and are often called the “fight-or-flight” reactions.

**Synthesis of Catecholamines**



### Degradation of Catecholamines:



VMA is increased with pheochromocytomas, tumors of the adrenal characterized by excessive production of catecholamines.

### ➤ MAO inhibitors

- MAO is found in neural and other tissues, such as the intestine and liver.

- MAO inhibitors may irreversibly or reversibly inactivate the enzyme, permitting neurotransmitter molecules to escape degradation and, therefore, to both accumulate within the presynaptic neuron and to leak into the synaptic space.
- This causes activation of norepinephrine and serotonin receptors, and may be responsible for the antidepressant action of these drugs.

### ➤ **Serotonin**

- Serotonin, also called 5-hydroxytryptamine (5HT), is synthesized and stored at several sites in the body.
- The largest amount of serotonin is found in cells of the intestinal mucosa. Smaller amounts occur in the central nervous system, where it functions as a neurotransmitter, and in platelets.
- Serotonin has multiple physiologic roles, including pain perception, regulation of sleep, appetite, temperature, blood pressure, cognitive functions, and mood (causes a feeling of well-being).
- Serotonin is converted to melatonin in the pineal gland via acetylation and methylation.

### **Synthesis of Serotonin**

